

693. GUSTAVIA LONGIFOLIA

Lecythidaceae

Ghilleen Prance and Scott Mori

Summary. *Gustavia longifolia* Poeppig ex Berg, a native of western Amazonia forest is described and illustrated.

The genus *Gustavia* was described in 1775 by Linnaeus in honour of King Gustav III of Sweden. Linnaeus certainly picked an attractive and interesting plant to honour his monarch. The genus was based on a single species *G. augusta* L. which is a white-flowered species that is the most widely distributed one in the genus. It is common along river margins throughout Amazonia and in the Guianas and the distribution extends as far into north-eastern Brazil as the State of Pernambuco. Linnaeus described *Gustavia* based on the collections of Frederik Allamand from the herbarium of Daniel Solander made between 1750 and 1770 in Suriname in the Guianas. Linnaeus said of this plant: '*arbor alta, floribus albis, magnis, spectatissima.*' He was obviously aware of the spectacular beauty of the flower of this species of *Gustavia*. He placed *Gustavia* in the group Monadelphia of his classification, but soon it was linked to other members of the Lecythidaceae or Brazil nut family, which was described as an entity in 1825 by Achille Richard. In the same work Linnaeus mentioned another species that is now in Lecythidaceae, *Lecythis ollaria* L. which he placed in his class Polyandra and in the genus described by his student Pehr Loeffling in 1758. *Gustavia* belongs to the part of the Lecythidaceae that has regular or actinomorphic flowers and is considered basal among the Neotropical genera. The Brazil nut (*Bertholletia excelsa* Bonpl.) and most other genera have zygomorphic flowers with an elaborate androecium structure (see Mori in Mori & Prance, 1979, Fig. 9 and Mori & Lepsch-Cunha, 1995, Figs. 1–3).

The whole genus *Gustavia* was monographed by Scott Mori in 1985 (Mori & Prance, 1979) who recognised 40 species at that time. *Gustavia longifolia* was first described by Otto Berg (1858) in his monograph of the Lecythidaceae for Martius' *Flora brasiliensis*. It was based on an 1830 collection of Eduard Friedrich Poeppig from the forests of Loreto Province in Peru. The most common local name for this species is



Plate 693 *Gustavia longifolia*

LUCY SMITH

chope or *sacha-chope*. Many synonyms exist especially from the work of Reinhardt Knuth (1939) who created many unnecessary names in the Lecythidaceae. The great Anglo-Brazilian artist Margaret Mee painted two species of *Gustavia*. These are *G. pulchra* Miers and *G. augusta* L. which are shown in their native habitat of the Amazon forest and were published in Mee (1988).

Gustavia longifolia recently flowered in the nursery at Kew, which gave Lucy Smith the opportunity to paint it. Carlos Magdalena of the nursery staff was able to make some observations and he noted that the flowers open at night. On the first night no pollen is released; then late on the next day the female stage is suppressed and the pollen is released. This is the first time that stigma receptivity and pollen release at different times has been observed in the Lecythidaceae. Carlos experimented with buzzing the stamens by using a tuning fork and this released the pollen only from second-day flowers. If this is not done or presumably if no insect visits to buzz the flower in the field then the pollen falls onto the receptacle as the flowers senesce. There are no field observations of the pollination of *G. longifolia*, but this observation in a cultivated species is interesting because Lecythidaceae specialist Scott Mori has observed the pollination of the related species *Gustavia augusta* L. in French Guiana. The genus is known to be bee pollinated and the flowers of *G. augusta* are visited by small *Trigona* bees in the day and by a larger bee at night. It is thought that *G. augusta* is pollinated by the night-flying species of bee, *Megalopta genalis*, that buzz pollinates the flowers. Since *G. longifolia* also flowers at night is probable that it is also pollinated by a night-flying bee.

CULTIVATION. At Kew, *Gustavia longifolia* is cultivated in a moist tropical glasshouse that it is maintained at a minimum of 20°C and above 70% relative humidity. Specimens are kept shaded at all times as leaves appear to scorch if exposed to direct sunlight, especially at midday in summer. In the tropical nursery, the species is grown in pots containing Kew's Number 3 potting medium, an organic multi-purpose compost with 10% of fine grade loam and 90% of other organic materials such as Silvafibre and coir. However, this and other species of *Gustavia* (such as *G. augusta* and *G. poeppigiana* O. Berg.) are also cultivated in glasshouse display beds that have more loam and sand in the substrate. Humidity and shading are important when cultivating *G. longifolia*, since it is an understorey

plant of rainforests and, in common with many other species that grow in similar habitats, tends to be a 'flush' grower: the growing tips burst with a number of small, pale brown, soft and somewhat floppy leaves which rapidly expand, turn green and become papery and robust. The plant then enters a period of rest, until a new flush of leaf growth appears; at Kew all flowering has occurred between two cycles of leaf development.

PROPAGATION. It is quite likely that *Gustavia* – in common with other Lecythidaceae, such as *Barringtonia* – can be propagated by cuttings, but this has not yet been tried. However, *Gustavia* species tend to grow small suckers near the base of the trunk, especially on leaning specimens, and these small suckers can be separated from the mother plant. Care is taken to obtain as many roots as possible. At Kew, these suckers are then treated as cuttings, potted in a mix of coir (70%) and perlite (30%) and placed in a misting/fogging unit. Once new roots appear, the suckers are potted in the Kew no. 3 compost but should be kept inside a closed case with extra humidity until well-established.

Despite repeat attempts to pollinate the flowers, not a single fruit has been produced from this species at Kew so far. This may suggest that *G. longifolia* is self-incompatible (though spontaneous seed set has been recorded in other species such as *G. poeppigiana*). Interestingly, *Gustavia* seeds have been reported to be able to form small plantlets on pieces of cotyledon. This will be an experiment to try if seed ever becomes available.

Gustavia longifolia Poeppig ex Berg, Martius Flora brasiliensis 14 (1): 472 (1858). Type: Peru, Loreto, sylvae obscurae ad yurimaguas, *Poeppig 2094* (lectotype, W).

Japarandiba longifolia (Poeppig ex Berg) O. Kuntze, Rev. Gen. 1: 240 (1885).

Japarandiba spruceana Ule in Karsten & Schenck. Vegetationsbilder Reihe 2, Heft 1, tafel 1 (1904).

Gustavia mangua Macbride, Publ. Field Mus. Nat. Hist., Bot. Ser. 11: 29 (1931).

Gustavia iquitosensis R. Knuth, Repert. Spec. Nov. Regni Veg. 35: 338 (1934).

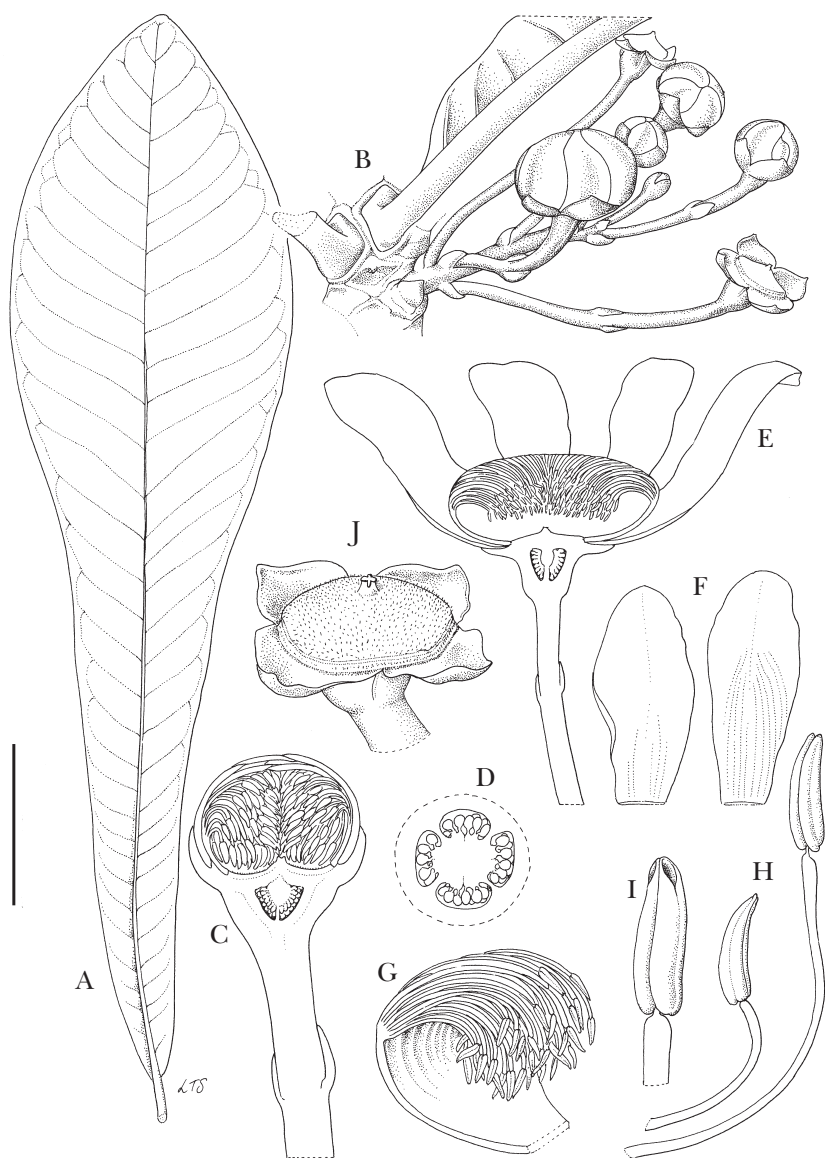
Gustavia tessmannii R. Knuth, Repert. Spec. Nov. Regni Veg. 35: 338 (1934).

Gustavia spruceana R. Knuth in Engler, Pflanzenreich IV. 219a: 27 (1939).

Gustavia duckei R. Knuth in Engler, Pflanzenreich IV. 219a: 28 (1939).

Gustavia magna Cuatrecasas, Fieldiana Bot. 27 (2): 93–94 (1951).

Gustavia dumitiana Garcia-Barriga, Mutisia 12: 4–7 (1952)



***Gustavia longifolia*.** A, leaf (see scale bar); B, inflorescence with buds, $\times \frac{3}{4}$; C, section of bud $\times 1.5$; D, section of ovary bud, $\times 4$; E, flower section $\times 1$; F, petals $\times \frac{3}{4}$; G, portion of androecium $\times 2$; H, short and long stamens $\times 5$; I, anther at anthesis, $\times 8$; J, ovary, $\times 8$. Scale bar: A = 10 cm; B, F, = 2.5 cm; C = 1.5 cm; D = 5 mm; E = 2 cm; G = 1 cm; H = 4 mm; I = 2.5 mm. Drawn by Lucy Smith from living material at Kew.

DESCRIPTION. *A tree* to 30 m tall and 50 cm diameter, the leaf bearing branches 7–20 mm in diameter, with leaves borne in one or more clusters at the end of the branches. *Bark* brown, nearly smooth. *Leaf blades* narrowly obovate or oblanceolate, 40–120 × 8–38 cm, glabrous or minutely puberulous on the abaxial veins, with 27–49 pairs of lateral veins, narrowly tapered to an acute or cuneate base; apices acute or acuminate; margins entire, minutely serrulate or serrate petioles 5–90 × 4–15 mm, usually semicircular in cross section. *Inflorescences* cauline, racemose, borne under the leaves, with 5–16 flowers, the rachis 15–70 mm; pedicels 30–70 mm, subtended by a single ovate to oblong, rounded basal bract 2–5–6 × 2–5 mm and bearing at or slightly below the middle 2 ovate, cucullate bracteoles 2–5 × 2–8 mm. *Flowers* 6–16.5 cm in diameter; calyx with four broadly triangular or shallowly rounded lobes, 2–4 × 6–18 mm; *petals* 8(–9), narrowly oblong to oblanceolate, 30–80 × 19–45 mm, densely pubescent in bud, puberulous at anthesis, pale purple to dark red, sometimes rose-pink; connate androecial base 6–20 mm high, the outermost filaments 12–25 mm, yellow with pale purple at the apex; *anthers* 2–5 mm; ovary often with four costae, usually rusty-tomentose on the exterior, 4(–6) locular, white-woolly at the summit; style short, 1–1.5 mm, conical; stigma with four lobes. Fruits poorly known, to 75 mm long. Seeds angular in cross section, 30 × 20 mm, without a distinct funicle.

DISTRIBUTION. Endemic to W Central Amazonia in Colombia Ecuador and Peru and as far east as the Rio Negro in Brazil.

HABITAT. An understory tree of the upland terra firme (non-flooded) forests.

FLOWERING TIME. In the rainy season mainly from October to February.

ACKNOWLEDGEMENTS. We are grateful to Carlos Magdalena for the account of the cultivation of this species at Kew and to Lucy Smith for the painting and drawings.

REFERENCES

- Berg, O.C. (1858). Myrtaceae subtribus II. Barringtonieae, subtribus I Lecythideae. In: Martius. *Flora brasiliensis* 14(1): 469–526.
- Knuth, R. (1939). Lecythidaceae. In: Engler, A., *Pflanzenreich IV* 219a: 1–146.
- Linnaeus, C. (1775). *Gustavia*. In: *Plantae Surinamenses*. Uppsala.
- Mee, M. (1988). In: Morrison, Tony (ed.). *In Search of Flowers of the Amazon Forests: Diaries of an English Artist Reveal the Beauty of the Vanishing Rainforest*. Nonesuch Expeditions, Woodbridge, Suffolk.
- Mori, S.A. & Lepsch-Cunha, N. (1995). The Lecythidaceae of a Central Amazonian moist forest. *Memoires of the New York Botanical Garden* 75: 1–55.
- Mori, S.A., Prance, G.T. & Mori S.A. (1979). Lecythidaceae Part I. The actinomorphic-flowered New World Lecythidaceae. *Flora Neotropica Monograph* 21: 1–270.
- Richard, A. (1825). In Bory de Saint-Vincent, J.B.G.M., *Dictionnaire classique d'histoire naturelle*, 9: 259.